

FDS EXPERIMENT-9

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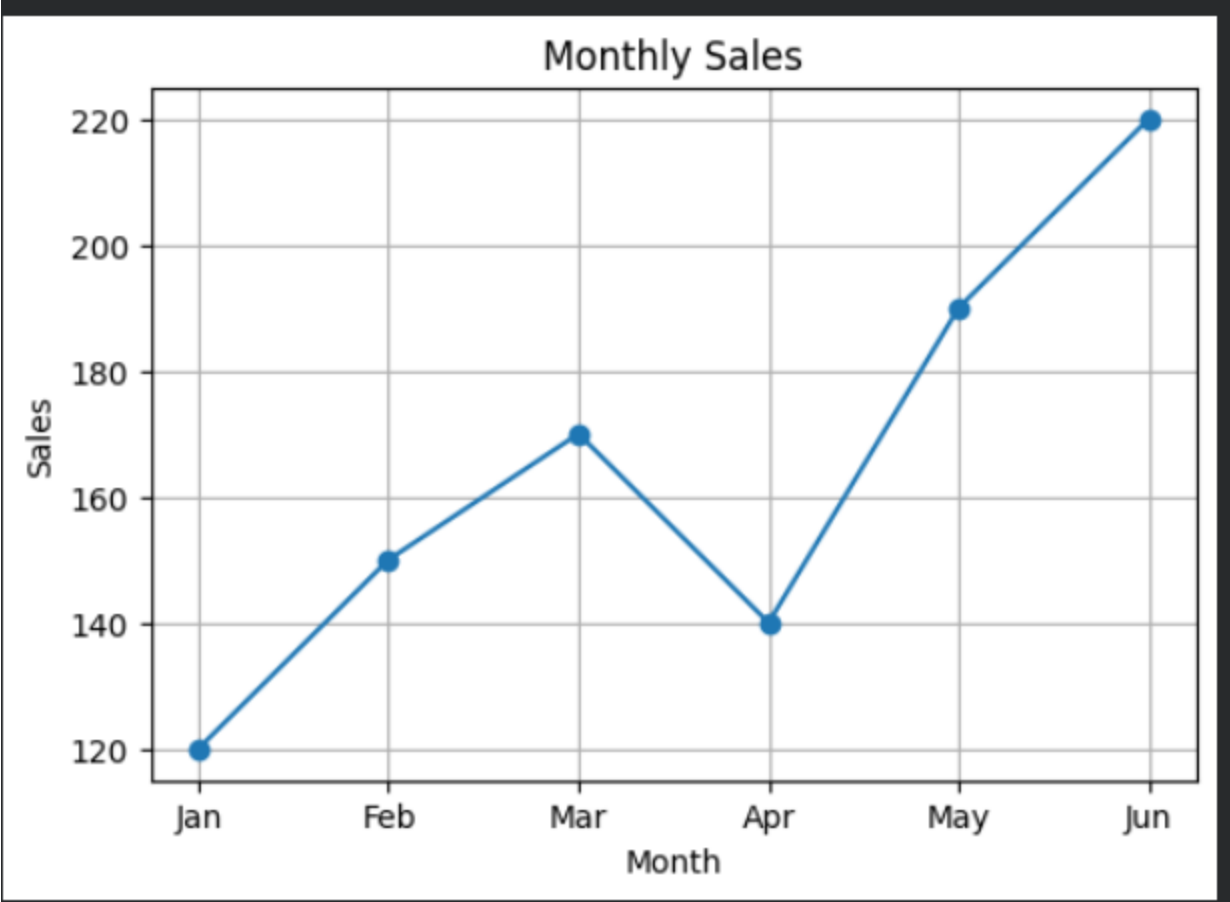
REG NO:250701697

Experiment to understand Matplotlib library use cases in Data Science through visualization.

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 170, 140, 190, 220]

# 1. Line Plot
plt.figure(figsize=(6, 4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



[]



3. Scatter Plot

```
advertising = [10, 15, 20, 25, 30, 35]
```

```
plt.figure(figsize=(6, 4))
```

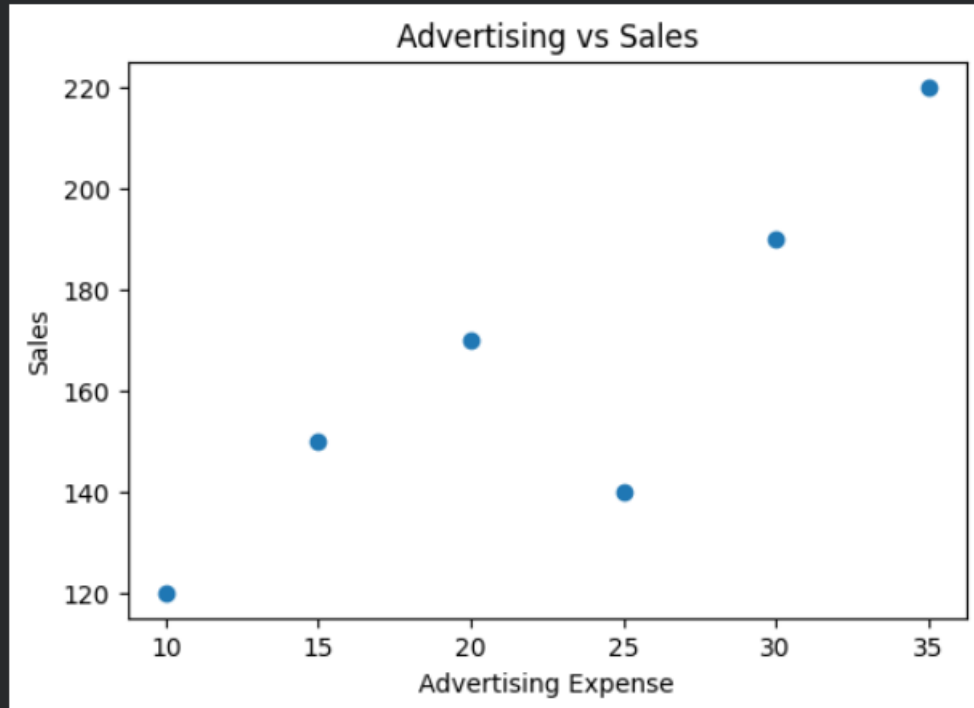
```
plt.scatter(advertising, sales)
```

```
plt.title("Advertising vs Sales")
```

```
plt.xlabel("Advertising Expense")
```

```
plt.ylabel("Sales")
```

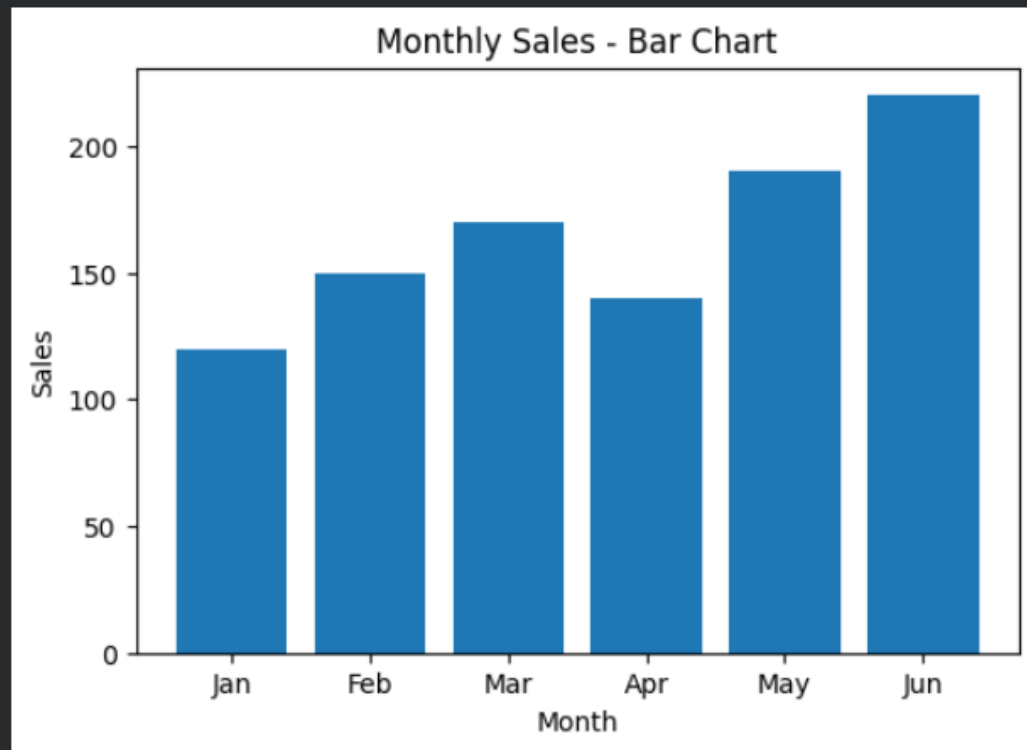
```
plt.show()
```





2. Bar Chart

```
plt.figure(figsize=(6, 4))  
plt.bar(months, sales)  
plt.title("Monthly Sales - Bar Chart")  
plt.xlabel("Month")  
plt.ylabel("Sales")  
plt.show()
```



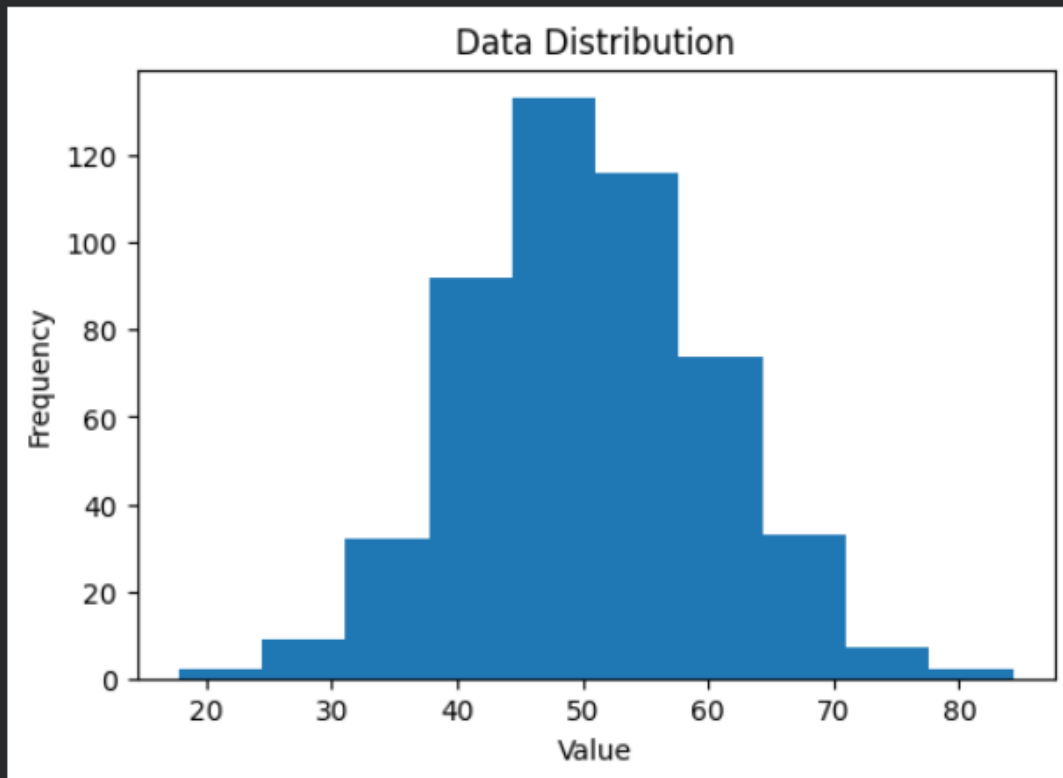


4. Histogram

```
data = np.random.normal(50, 10, 500)
```

```
plt.figure(figsize=(6, 4))  
plt.hist(data, bins=10)  
plt.title("Data Distribution")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```

...



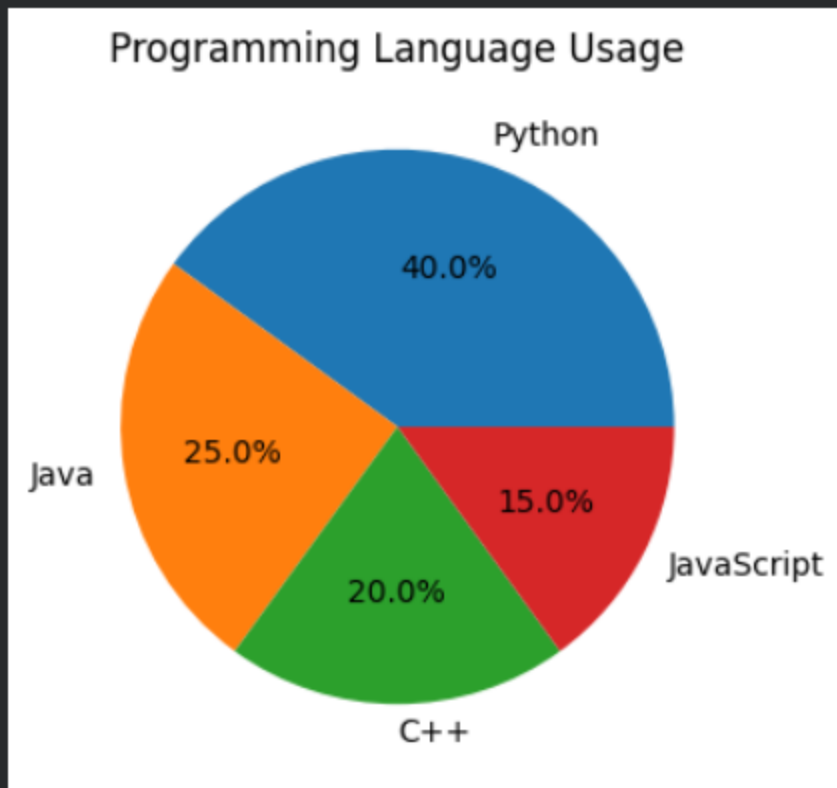


5. Pie Chart

```
labels = ['Python', 'Java', 'C++', 'JavaScript']  
values = [40, 25, 20, 15]
```

```
plt.figure(figsize=(6, 4))  
plt.pie(values, labels=labels, autopct='%1.1f%%')  
plt.title("Programming Language Usage")  
plt.show()
```

...



```
# 6. Box Plot
data = [12, 15, 14, 10, 18, 20, 16, 15, 14, 50]

plt.figure(figsize=(6, 4))
plt.boxplot(data)
plt.title("Box Plot")
plt.ylabel("Values")
plt.show()
```

